



HR Attrition Analysis Dashboard

Diagnosing Employee Turnover Through Data

An end-to-end HR analytics project using **SQL**, **Power BI** & **Excel**

PEOPLE ANALYTICS

DATA-DRIVEN RETENTION

Sandip Mandal

Business Context

Unmanaged attrition quietly erodes organizational value — through recruitment costs, lost productivity, knowledge drain, and disrupted team performance. This project was built for the **HR / People Analytics function** to shift from reactive reporting to **proactive, data-backed retention strategy**.

Scale

10,000-employee workforce dataset analyzed across departments, roles, seniority, and work conditions

Gap

HR relied on fragmented, manual reports — no single source of truth to guide retention decisions

Goal

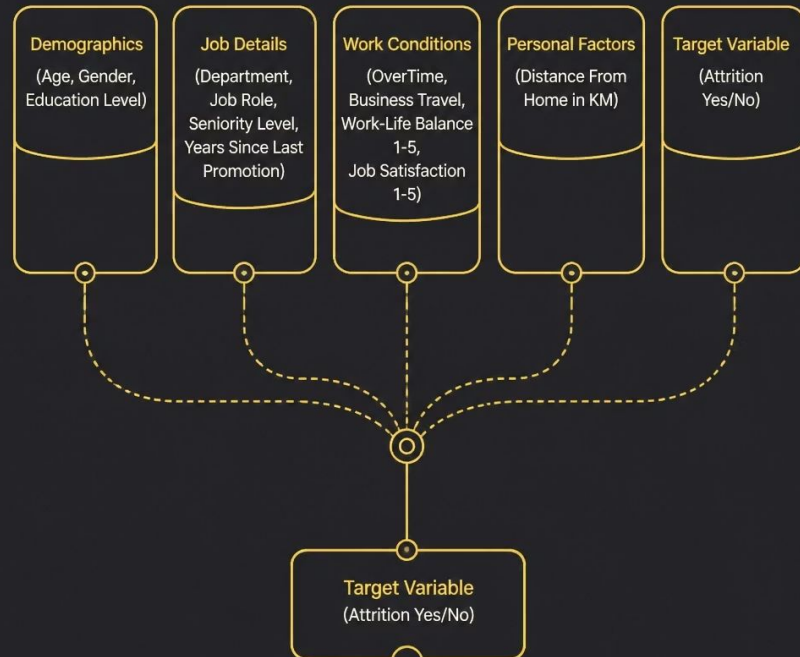
Uncover attrition patterns and convert raw data into actionable, narrative-driven insights



Sandip Mandal

About the Data

Dataset: **10,000 employee records** structured like enterprise HRIS exports — cleaned, validated, and modeled for attrition analysis.



Data Coverage

- **Demographics:** Age, Gender, Education Level
- **Job Details:** Department, Role, Seniority, Years Since Last Promotion
- **Work Conditions:** Overtime, Travel, Work-Life Balance, Job Satisfaction
- **Personal Factors:** Distance From Home to Office (KM)
- **Target Variable:** Attrition (Yes / No)

The Central Question

"Who is leaving the company, and why?"

1

Departments & Roles

Which departments and job roles have the highest attrition rates?

2

Commute Distance

Are attrition rates linked to distance from home to office?

3

Career Growth

Does stalled promotion drive employees to leave?

4

Seniority Risk

Which seniority level is most at risk of leaving?

5

Workload Factors

Do overtime, travel demands, and education level correlate with attrition?

6

Satisfaction Signals

Are work-life balance and job satisfaction measurable predictors of turnover?

Project Objectives & Methodology

Build an **interactive, executive-ready dashboard** that consolidates all attrition drivers — enabling HR leadership to filter, drill down, and act in real time.



SQL

Extracted, cleaned, and queried raw HR data. Built aggregations for attrition rate, department counts, and seniority breakdowns. Performed validation and de-duplication before loading into Power BI.



Excel

Used for exploratory data analysis, pivot-table validation, and pre-modeling checks. Cross-verified KPI calculations — attrition %, average age, department totals — before dashboard build.



Power BI

Built fully interactive dashboard with DAX measures for KPIs. Designed drill-down navigation across 3 pages (Overview, Distance Analysis, Role & Department). Added Gender and Department slicers and narrative insight boxes.

Dashboard Overview: Key Metrics

10K

Total Employees

7,753

Active Employees

2,247

Attrition Employees

22.47%

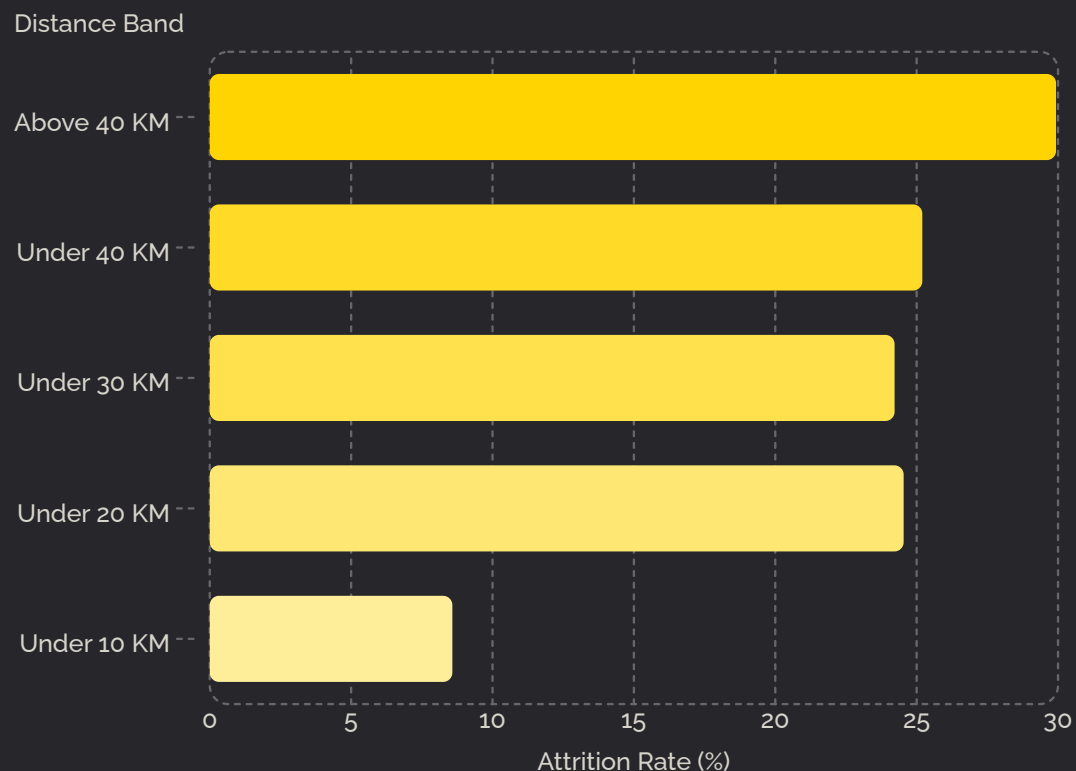
Attrition Rate

40

Average Age

 **Critical Signal:** A 22.47% attrition rate represents a material retention risk — requiring immediate root-cause investigation and targeted intervention across departments and roles.

Insight 1: Attrition by Distance From Home



Key Takeaway

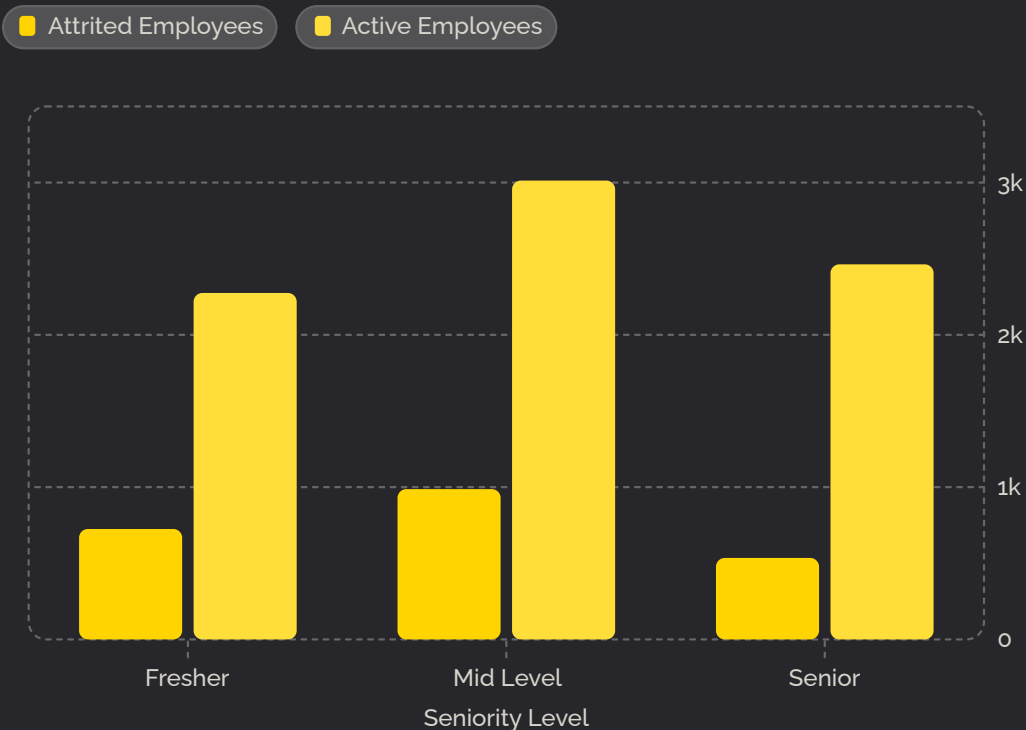
Employees living farther from the office attrite at nearly **3.5x the rate** of those living close by.

Commute burden is one of the strongest predictors of turnover in this dataset. The sharp drop at Under 10 KM (8.58%) vs. Above 40 KM (29.93%) signals a clear geographic retention risk.



Action Implication: Remote/hybrid work policy and commute support programs could materially reduce attrition.

Insight 2: Attrition by Seniority Level



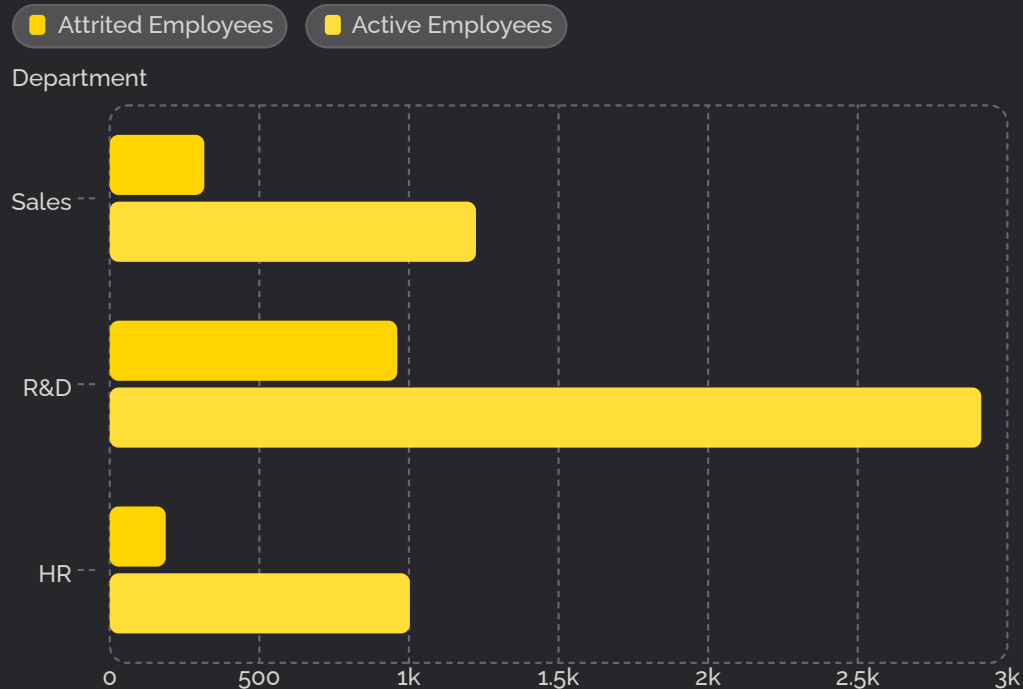
Key Takeaway

Mid-level employees have the highest absolute attrition count (987), but Freshers show the highest attrition rate at 24.17% — nearly 1 in 4 new hires leaves.

Seniors attrite at the lowest rate (17.83%), suggesting stronger organizational commitment at higher levels.

Action Implication: Strengthen onboarding, early-career development, and 90-day check-ins to improve Fresher retention.

Insight 3: Attrition by Department



Key Takeaway

R&D has the highest absolute attrition (961 employees), but Sales has the highest attrition rate at 20.52% — the most vulnerable department by proportion.

HR has the lowest attrition rate at 15.71%, suggesting stronger internal retention in the People function itself.

📌 **Action Implication:** Sales requires targeted retention programs; R&D needs role-specific exit analysis to address high-volume attrition.



Conclusion & Recommendations

This project demonstrates how **SQL, Power BI, and Excel** can transform raw HR data into a strategic retention tool — enabling HR leaders to move from reactive reporting to proactive, data-backed decision-making.



Remote/Hybrid Policy

Reduce commute burden — the strongest attrition predictor — through flexible work arrangements



Early-Career Programs

Strengthen onboarding and 90-day check-ins to address the 24.17% Fresher attrition rate



Department Interventions

Launch targeted retention programs in Sales (20.52% rate) and R&D (highest volume)